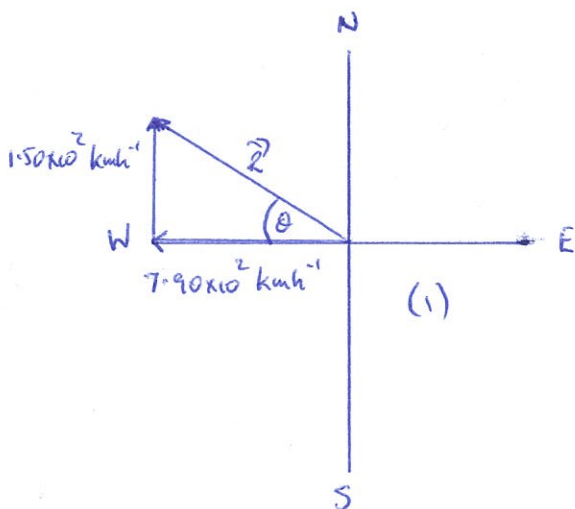


**YEAR 10 SCIENCE EXTENSION
PHYSICS
ASSIGNMENT 1**

NAME: SOLUTIONSMARK: 45

1. A plane is flying at an altitude of 10,000 m at $7.90 \times 10^2 \text{ kmh}^{-1}$ W. There is a strong wind blowing from the south at $1.50 \times 10^2 \text{ kmh}^{-1}$. Determine the plane's velocity over the ground. Include a vector diagram.



$$\vec{R} = \sqrt{(7.90 \times 10^2)^2 + (1.50 \times 10^2)^2}$$

$$= 8.041 \times 10^2 \text{ kmh}^{-1} \quad (1)$$

$$\tan \theta = \frac{1.50 \times 10^2}{7.90 \times 10^2}$$

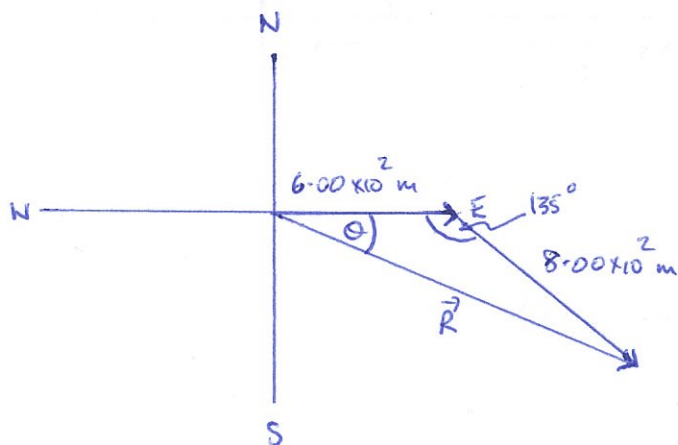
$$\Rightarrow \theta = 10.75^\circ \quad (1)$$

$$\therefore \text{Velocity} = 8.04 \times 10^2 \text{ kmh}^{-1} \text{ W } 10.8^\circ \text{ N} \quad (1)$$

(4)

2. An athlete runs $6.00 \times 10^2 \text{ m}$ E in 95.0 s, then $8.00 \times 10^2 \text{ m}$ SE in 2.00 minutes.

(a) Calculate her displacement.



$$\vec{R} = \sqrt{(6.00 \times 10^2)^2 + (8.00 \times 10^2)^2 - 2(6.00 \times 10^2)(8.00 \times 10^2) \cos 135^\circ}$$

$$= 1.296 \times 10^3 \text{ m} \quad (1)$$

$$\frac{8.00 \times 10^2}{\sin \theta} = \frac{1.296 \times 10^3}{\sin 135^\circ}$$

$$\Rightarrow \theta = 25.88^\circ \quad (1)$$

$$\therefore \text{Displacement} = 1.30 \times 10^3 \text{ m E } 25.9^\circ \text{ S} \quad (1)$$

(3)

- (b) Determine her average speed.

$$\begin{aligned}\text{Speed} &= \frac{d}{t} \\ &= \frac{(6.00 \times 10^2 + 8.00 \times 10^2)}{(95.0 + 120.0)} \quad (1) \\ &= 6.512 \text{ ms}^{-1}\end{aligned}$$

$$\therefore \text{Average speed} = 6.51 \text{ ms}^{-1} \quad (1)$$

(2)

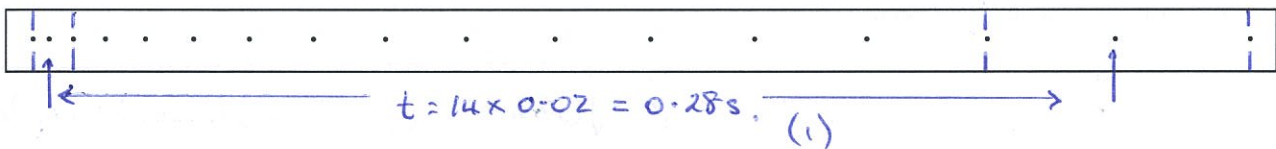
- (c) What is the athlete's average velocity?

$$\begin{aligned}V_{\text{ave}} &= \frac{S}{t} \\ &= \frac{1.296 \times 10^3}{(95.0 + 120.0)} \quad (1) \\ &= 6.028 \text{ ms}^{-1} \quad (1)\end{aligned}$$

$$\therefore V_{\text{ave}} = 6.03 \text{ ms}^{-1} \text{ E } 25.9^\circ \text{ S} \quad (1)$$

(3)

3. In recording the motion of a toy car down an inclined track, a student produced the following tape using a ticker timer operating at 50 dots per second. Analyse the tape and determine the acceleration of the car. (Use the diagram below.)



$$\begin{aligned}u &= \frac{s}{t} \\ &= \frac{5.0 \times 10^{-3}}{0.04} \\ &= 0.12 \text{ ms}^{-1} \quad (1)\end{aligned}$$

$$\begin{aligned}v &= \frac{s}{t} \\ &= \frac{35.0 \times 10^{-3}}{0.04} \\ &= 0.875 \text{ ms}^{-1} \quad (1)\end{aligned}$$

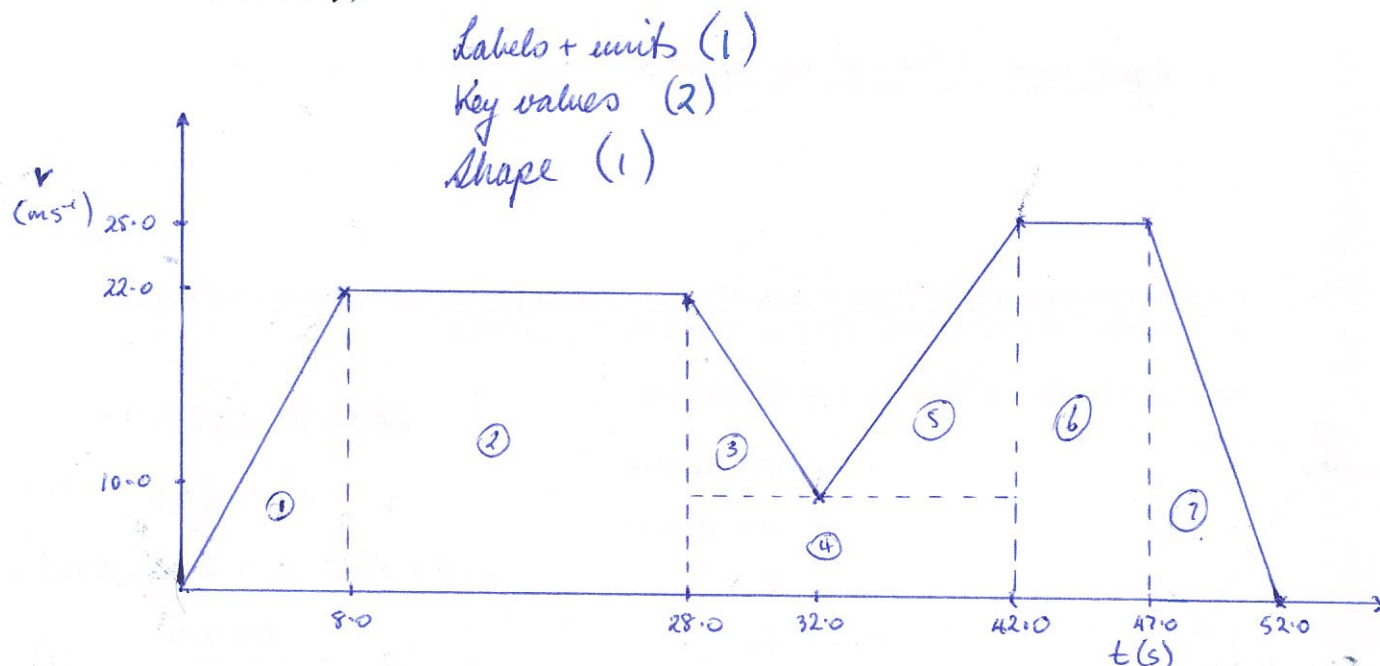
$$\begin{aligned}a &= \frac{v-u}{t} \\ &= \frac{0.875 - 0.12}{0.28} \\ &= 2.70 \text{ ms}^{-2} \quad (1)\end{aligned}$$

(4)

4. A car undergoes the following motion between three sets of traffic lights A, B and C.

The car accelerates uniformly from rest at light A and reaches 22.0 ms^{-1} after 8.00 s . It maintains this velocity for 20.0 s before slowing uniformly as it approaches lights B. The lights turn green so the car, having reduced its velocity to 10.0 ms^{-1} in 4.00 s , accelerates uniformly for 10.0 s to reach 25.0 ms^{-1} . It maintains this velocity for 5.00 s before uniformly decelerating to a stop in 5.00 s at lights C.

- (a) Draw a velocity-time graph for the motion. (The scales only need to show the important data and need not run evenly.)



(4)

- (b) Using the graph (or otherwise), calculate the size of the decelerations that occurred during the motion.

$$a_3 = \frac{(10.0 - 22.0)}{(32.0 - 28.0)} \quad (1)$$

$$= -3.00 \text{ ms}^{-2} \text{ backwards} \quad (1)$$

$$a_7 = \frac{(0 - 25.0)}{(52.0 - 47.0)} \quad (1)$$

$$= -5.00 \text{ ms}^{-2} \text{ backwards} \quad (1)$$

(4)

- (c) How far apart are lights A and C?

$S = \text{area under graph}$

$$= \frac{1}{2}(8.00)(22.0) + (20.0)(22.0) + \frac{1}{2}(4.0)(12.0) + (14.0)(10.0) + \frac{1}{2}(10.0)(15.0) + (5.0)(25.0) + \frac{1}{2}(5.0)(25.0) \quad (3)$$

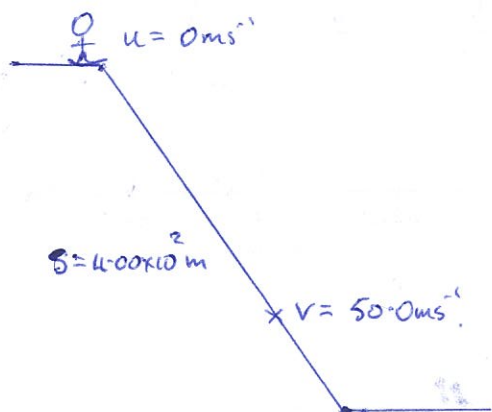
$$= 9.545 \times 10^2 \text{ m} \quad (1)$$

$\therefore$ lights are $9.54 \times 10^2 \text{ m}$ apart.

(4)

5. A downhill skier of mass 80.0 kg started from rest on top of a mountain and accelerated smoothly down a uniform slope. After moving $4.00 \times 10^2 \text{ m}$ his velocity was $1.80 \times 10^2 \text{ kmh}^{-1}$.

- (a) Calculate the acceleration experienced by the skier.



$$v = 50.0 \text{ ms}^{-1}$$

$$u = 0.0 \text{ ms}^{-1}$$

$$a = ?$$

$$t = ?$$

$$s = 4.00 \times 10^2 \text{ m}$$

down the slope is +ve.

$$v^2 = u^2 + 2as \quad (1)$$

$$\Rightarrow (50.0)^2 = 0 + 2a(4.00 \times 10^2) \quad (1)$$

$$\Rightarrow a = \frac{2.50 \times 10^3}{8.00 \times 10^2}$$

$$= 3.125 \text{ ms}^{-2}$$

$\therefore$ Acceleration = 3.12 ms^{-2} down the slope. (1)

(3)

- (b) How long did it take for him to travel this far?

$$v = u + at \quad (1)$$

$$\Rightarrow 50.0 = 0 + (3.125)t \quad (1)$$

$$\Rightarrow t = \frac{50.0}{3.125}$$

$$= 16.0 \text{ s} \quad (1)$$

(3)

- (c) Would the skier actually have the velocity mentioned in the question? Explain your answer.

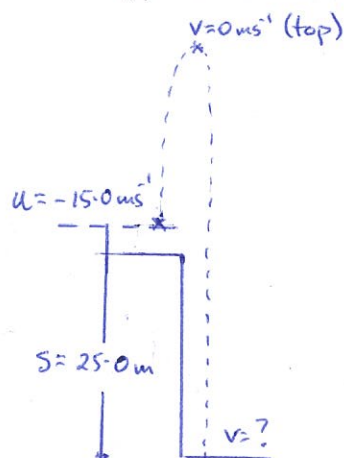
no. (1)

Air resistance and friction with the snow will slow the skier. (1)

(2)

6. A boy throws a stone upwards at 15.0 ms^{-1} from the edge of a cliff 25.0 m high. It just misses the edge on the way down and falls to the bottom of the cliff.

- (a) Calculate the maximum height reached by the stone above the ground.



Consider only movement to the top.

↓ +ve. $v = 0 \text{ ms}^{-1}$
 $u = -15.0 \text{ ms}^{-1}$
 $a = 9.80 \text{ ms}^{-2}$
 $t = ?$
 $s = ?$

$$v^2 = u^2 + 2as$$

$$\Rightarrow 0 = (-15.0)^2 + 2(9.80)s \quad (1)$$

$$\Rightarrow s = \frac{-225.0}{19.6}$$

$$= -11.48 \text{ m.} \quad (1)$$

Height = 36.5 m above the ground. (1)

(3)

- (b) With what velocity does the stone strike the ground?

Take whole motion. ↓ +ve.

$v = ?$
 $u = -15.0 \text{ ms}^{-1}$
 $a = 9.80 \text{ ms}^{-2}$
 $t = ?$
 $s = 25.0 \text{ m}$

$$v^2 = u^2 + 2as \quad (1)$$

$$= (-15.0)^2 + 2(9.80)(25.0) \quad (1)$$

$$= 715.0$$

$$\Rightarrow \underline{v = 26.7 \text{ ms}^{-1} \text{ down.} \quad (1)}$$

(3)

- (c) How long is the stone in the air?

$$v = u + at \quad (1)$$

$$\Rightarrow 26.7 = -15.0 + 9.80t \quad (1)$$

$$\Rightarrow t = \frac{(26.7 + 15.0)}{9.80}$$

$$= \underline{4.25 \text{ s.} \quad (1)}$$

(3)

